Q.1. Write a function “uniqueE” which will take a list of numbers as input and return another list of unique elements in sorted order.

- Example:

Input: L = [3, 2, 2, 1]

output: L1 = [1, 3]

```
# Prog takes a list and returns all the unique elements in a sorted list

def uniqueE(L):

   n=len(L)

   M=[]

   for i in range(n):

      c=L.count(L[i])

      if c==1:

         M.append(L[i])

   M.sort()

   return(M)

L=[3,2,2,1]

M=uniqueE(L)

print(M)
```

Q.2. Given a List L containing integers, Write a function that creates a dictionary containing all the numbers that occur twice or more in the list as keys and their indexes as values.

- Example:
- Input: L = [2, 2, 2, 1, 1, 1]
- Output: D = {2: [0, 1, 2], 1: [3, 4, 5]}

```
# Prog
def freq(L):
    dic={}
    for i in range(len(L)):
        item=L[i]
        f=L.count(item)
        if f>1:
            dic[item]=f

    return(dic)

L=[2,2,3,1,1,2]
print(L)
d=freq(L)
for i in d.keys():
    e=[]
    for index, elem in enumerate(L):
        if i==elem:
            e.append(index)
    d[i]=e
print(d)
```

Q.4. Write a program to create “Jumbled word game”

- Minimum two players
- Each player’s turn comes alternately
- Computer shows a jumbled word, the player need to guess the word, put a timer (5 sec)
- Correct guess fetches 1 point
- Incorrect guess fetches 0 point

```
# Jumbled Word Game

import random

def choose():


words=["rainbow","computer","science","programming","mathematics","player","condition","reverse","water","b
oard"]

    pick=random.choice(words)

    return pick

def jumble(word):

    jumbled="".join(random.sample(word,len(word)))

    return jumbled

def thank(p1n,p2n,p1,p2):

    print(p1n,"Your score is: ",p1)

    print(p2n,"Your score is: ",p2)

    print("Thanks for playing")

    print("Have a nice day")



def play():

    p1name=input("Player-1, please enter your name: ")

    p2name=input("Player-2, please enter your name: ")

    pp1=0 #player-1 points

    pp2=0 #player-2 points

    turn=0

    while(1):

        # Computer's task

        picked_word=choose()

        # Create question

        qn=jumble(picked_word)

        print(qn)
```

```
if turn%2==0:  # it is the turn for the player-1

    print(p1name,"Your turn")

    ans=input("What is on my mind? ")

    if ans==picked_word:

        pp1=pp1+1

        print("Your current score is: ",pp1)

    else:

        print("Better luck next time. I thought: ",picked_word)

    c=int(input("Press 1 to continue and 0 to quit: "))

    if c==0:

        thank(p1name,p2name,pp1,pp2)

        break


# it is the turn for the player-2

else:

    print(p2name,"Your turn")

    ans=input("What is on my mind? ")

    if ans==picked_word:

        pp2=pp2+1

        print("Your current score is: ",pp2)

    else:

        print("Better luck next time. I thought: ",picked_word)

    c=int(input("Press 1 to continue and 0 to quit: "))

    if c==0:

        thank(p1name,p2name,pp1,pp2)

        break

turn=turn+1
```

Lets Play the Game

play()

Q.5. What is L1 after the function call?

```
def mystery(L):
        L = L[2:]
        return L
L1 = [7, 11, 13, 17, 19, 21]
mystery(L1)
```

Q.6. What is L1 after the function call?

```
def mystery(L):
        L[0] = 5
        return L
L1 = [7, 11, 13, 17, 19, 21]
mystery(L1)
```

Q.7. Write a python function that takes a List of integers and returns True if the absolute difference between each adjacent pair of elements strictly decrease.

Example:

L = [9, 2, 7, 3, 1]

True

```
# Program Contracting List
def contracting(L):
  n=len(L)
  Ld=[]
  i=0
  while i<n-1:
```

```
        d=abs(L[i]-L[i+1])
        Ld.append(d)
        i=i+1
    for i in range(len(Ld)-1):
        if Ld[i]<=Ld[i+1]:
            return(False)
    return(True)
L=[9,2,7,3,1]
val=contracting(L)
print(val)
```